Instructional Strategies for Responsible GenAI Use in Computer Science Education

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Abstract

The emergence of commercial Generative AI (GenAI) tools like ChatGPT has led to a large body of research investigating how these tools are used in postsecondary CS education (CSEd) and their impact on student learning. GenAI offers students personalized and timely academic support, but its use also raises concerns about over-reliance and impacts on critical thinking skills. Instructors have therefore taken many different approaches toward the use of GenAI in their CS courses. Some instructors have limited the integration of GenAI in their courses through bans or restrictive assignments. Others have taken more proactive approaches that encourage using GenAI in a responsible way to support learning, facilitated by custom GenAI tools with pedagogical guardrails or instructor-provided guidance on how to best use commercial GenAI tools. My research is focused on instructional strategies for guiding students on how to use GenAI in CS courses; I have completed a Systematic Literature Review (SLR) on custom GenAI tools and plan to conduct another SLR on instructor-provided guidance on commercial GenAI use. The findings from both SLRs will be synthesized to understand best practices for teaching students how to use GenAI in a way that supports their learning. From this work I plan to develop instructional interventions that teach responsible GenAI use, iteratively refining the interventions through user studies and later evaluating them empirically through in-class deployments.

CCS Concepts

• **Social and professional topics → Computing education**.

Keywords

computer science education, generative AI, instructional strategies, instructor guidance, responsible AI use

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1 Introduction

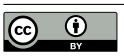
The rapid adoption of commercial GenAI tools like ChatGPT [23], Claude [2], and Microsoft Copilot [22] has resulted in a large body of research investigating how these tools are used in the context of CSEd and their impact on student learning [3, 4, 6, 10, 12, 13, 18, 26, 27]. GenAI tools can now easily solve most programming problems at the introductory level, presenting unique challenges for instructors who must decide how GenAI should be used in their courses [5, 17, 30]. There are several competing interests when it comes to adopting GenAI in CS coursework [25]. Students report valuing the problem-solving support [6], personalized explanations and examples [10], and timely, always-available feedback [6, 26] from GenAI. In contrast, instructors want to ensure they are preparing students to evaluate and debug AI-generated code and leverage GenAI in advanced CS work [10, 15, 21, 25, 31]. Nonetheless, students and instructors share several perspectives on GenAI, including concerns about over-reliance [9, 31] and impacts on problem-solving, metacognitive, and critical thinking skills [10, 28, 29].

The lack of consensus surrounding how to approach commercial GenAI tools in postsecondary CS coursework [7, 16, 25, 26] has left instructors to develop their own strategies for regulating student-AI interactions. Some instructors have opted to ban the use of GenAI in their coursework [16] or "AI-proof" their assessments [8], but these strategies may not be sustainable given the difficulty of enforcing bans [16] and potential impacts on student-instructor relationships [8]. Others have designed materials for students to self-govern their interactions with GenAI tools, such as through "personal contracts" for students to describe how they want to incorporate GenAI into their learning [24].

Another strategy for supporting student learning with GenAI is deploying custom GenAI tools that are trained on instructors' curricula [1, 20] or designed to avoid giving away coding solutions [14, 19], the latter a behavior desired by both students and instructors [6]. While these tools have the potential to promote the development of metacognition and self-regulation in students [29], Denny *et al.* [6] and Kapoor *et al.* [11] caution that a custom tool with too many constraints or restrictions may encourage students to prefer commercial GenAI tools without guardrails and thus hurt their learning.

Prather *et al.* [26] found that studies providing students with guidance on either custom tools with educational guardrails or commercial GenAI tools reported more positive outcomes than those providing unguided access, with no difference in student success between the two guided approaches. This finding is significant given that developing and maintaining a custom GenAI tool is not a practical approach for all instructors. Further, emphasizing responsible use of commercial tools may help students develop the skills needed to use GenAI effectively for advanced CS courses and future jobs [16, 26].

There is a need to characterize how CS instructors currently prepare their students to learn with GenAI, identifying potential gaps



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in their strategies—whether that be restrictions on GenAI use, deployment of custom tools, or guidance on using commercial tools—and understanding how different strategies impact student learning. In my prior work, I conducted an SLR of GenAI-powered CSEd tools to better understand tool development approaches, grounding in educational theories and framings, and evaluations done to empirically measure the impact of tools. The next step is to conduct a similar SLR on instructor-provided guidance on how to use commercial GenAI tools responsibly, understanding when, where, and how students receive guidance and examining the resulting outcomes on students. The strategies identified in both SLRs will inform my broader research goal of identifying best practices for instructional guidance, with a focus on introductory-sequence CS courses where students learn foundational problem-solving skills through programming. I plan to build on these findings to design and evaluate targeted instructional interventions that will help students use GenAI in a responsible way that supports rather than undermines their learning.

My work is guided by the following research questions:

(RQ1) How are students currently being taught to use commercial GenAI tools in a way that supports their learning?
(RQ2) How does structured guidance on commercial GenAI tools compare to other instructional strategies, such as the deployment of pedagogically oriented custom tools?
(RQ3) What practices best promote responsible GenAI use in CS, especially for introductory-sequence courses?

2 Research Plan

2.1 Prior Work

My prior work includes conducting an SLR in collaboration with other graduate student researchers on the use of custom GenAI tools for CS classrooms. These tools offer a pedagogically oriented alternative to commercial GenAI tools like ChatGPT or Claude, and also give instructors greater control over the amount of scaffolding provided to students by GenAI. We focused our SLR on custom tools in the postsecondary CSEd space, analyzing 74 papers and reporting our findings in a paper accepted to Koli Calling '26. Through inductive coding, we found that tool development is strongly motivated by desires to provide personalized learning experiences for students and to address the potential risks of commercial GenAI tools. Further, we found that tool development was often grounded in educational framings like scaffolding and cognitive load theory. Tools frequently included features to avoid giving direct answers to students and adapt to their responses over time. In terms of evaluation, the most common methods included post-surveys and logging usage data. From our results we also discussed broader trends we observed across the papers.

2.2 Current Work

To develop a more comprehensive understanding of how students are taught to use GenAI in CS coursework, I want to investigate how instructors provide guidance on using commercial GenAI tools, comparing this to my findings on custom tools and informing best practices for teaching responsible GenAI use. In this context, the "guidance" refers to specific interventions carried out by instructors to teach their students how to use GenAI; this could include lecture content discussing GenAI, a structured activity where students learn about productive or unproductive patterns of use, or an assignment outside of class that engages students to think critically about their use of GenAI. This work will not cover instructional strategies that are only focused on reflection—such as policies requiring students to explain their use of GenAI on coursework—instead emphasizing the role that instructors play in setting expectations on GenAI use in their classroom and how this impacts student learning. Similar to my prior work, I plan to conduct another SLR on this topic, following the same methodology of applying search queries to academic databases, screening and reading papers, extracting information with a codebook, and analyzing the results. I am interested in understanding the overarching themes on instructor guidance, as well as the format in which the guidance is delivered and to what extent the impact of the guidance is measured empirically, such as through analysis of students' academic performance or reported GenAI use. I hope to compare these findings with the results of my prior SLR on custom GenAI tools to identify shared themes in how students are taught to use GenAI.

Once both SLRs are completed, I want to identify best practices that can be delivered to students in introductory-sequence CS courses through instructional interventions. I anticipate there to be overlapping strategies that are effective with both custom and commercial GenAI tools, and hope to identify recurring strategies that are grounded in established theories of learning and education and have been rigorously evaluated. I will use these findings to develop instructional interventions designed to teach responsible GenAI use, which could take the form of case studies, worksheets, programming problem sets, or other materials that can be deployed in an introductory-sequence CS course. I plan to iteratively refine these interventions through user studies with students and eventually evaluate their effectiveness through in-class deployments.

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